

METUCHEN PUBLIC SCHOOLS

METUCHEN, N.J.



PRECALCULUS & HONORS PRECALCULUS (GRADES 10-12)

CURRICULUM

August 2024

Approved by the Metuchen School District Board of Education

August 27, 2024

METUCHEN PUBLIC SCHOOLS**DISTRICT ADMINISTRATION**

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CONTENT SUPERVISOR

Kate Lemerich, Secondary STEM Supervisor

CURRICULUM WRITERS

Chad Murray, Secondary Math Teacher

UNITS AT A GLANCE		
Unit #	Title	Suggested Time Frame / Pacing
I	Functions and Graphs	10 - 14 days
II	Functions: Polynomial, Rational, Exponential and Logarithmic	24 - 30 days
III	Trigonometry and Related Topics	65 - 75 days
IV	Sequences, Series, and Probability	20 - 30 days
V	Conics, Parametrics, and Polar Equations	10 - 14 days
VI	Introduction to Calculus: Limits & Derivatives	20 - 35 days

Honors Precalculus level work and/or pacing will be indicated with an asterisk (*) throughout the curriculum document.

UNIT OVERVIEW	
Unit # and Title: I. Functions and Graphs	Suggested Time Frame / Pacing: 10 - 14 days
Essential Questions: How do you find the slope of a line and use it to write an equation of the line? What are the important defining characteristics and representations of a function? How is the graph of a function used to determine the key elements of that function? How do you combine two functions to form a new function? What is the inverse of a function, and how do you represent it graphically and algebraically?	
NJSLS Content Standards: MA.F-BF.B.3 - Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$ and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. MA.F-BF.B.4b - Verify by composition that one function is the inverse of another. MA.F-BF.B.4c - Read values of an inverse function from a graph or a table, given that the function has an inverse. MA.F-IF.B.4 - For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity. MA.F-IF.B.5 - Relate the domain of a function to its graph and, where applicable, to the quantitative relationship it describes. For example, if the function gives the number of person-hours it takes to assemble n engines in a factory, then the positive integers would be an appropriate domain for the function. MA.F-IF.C.7a - Graph linear and quadratic functions and show intercepts, maxima, and minima. MA.F-IF.C.7b - Graph square root, cube root, and piecewise-defined functions, including step functions and absolute value functions. MA.F-IF.C.7c - Graph polynomial functions, identifying zeros when suitable factorizations are available, and showing end behavior.	

MA.F-IF.C.8a - Use the process of factoring and completing the square in a quadratic function to show zeros, extreme values, and symmetry of the graph, and interpret these in terms of a context.

Unit Content Vocabulary:

Distance Formula	Midpoint Formula	Parallel/Perpendicular lines	Slope
Increasing/Decreasing	Zeros/Intercepts	Domain/Range	Symmetry
Even/Odd Function	Linear function	Square/Cube Root Function	Asymptote
Absolute Value Function	Quadratic/Cubic Function	Vertical/Horizontal Shifts	Reflection in x/y axes
Transformations	Composition of Functions	Inverse Function	One-to-one
Horizontal Line Test			

General Resources: Textbook, Graphing Calculators, Desmos, DeltaMath, Khan Academy, Youtube videos, Kuta Software, Math Medic, Quizizz, Kahoot!, Citizen Math, Which One Doesn't Belong? (wodb), Open Middle, GeoGebra

Topic and Related Content: <i>What information do students need to know, by topic?</i>	Objectives: <i>Students will be able to:</i>	Suggested Resources, Core and Supplemental Materials and Activities, Technology resources <i>(Include Varied Levels of Text)</i>
Lines in the Plane	• Use a coordinate plane to model and solve real-life problems • Sketch graphs of equations; find x and y intercepts of graphs of equations • Use graphs of equations in solving real-life problems • Use slope to graph and write linear equations in two variables • Use slope to identify parallel and perpendicular lines	Section 1.1 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Functions	• Identify and evaluate functions • Analyze graphs of functions • Determine Domain & Range of a function	Section 1.2 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice

	Describe inequalities with interval notation	DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Graphs of Functions	- Describe intervals where a function is increasing, decreasing, or constant. - Determine, algebraically and graphically, whether a function is odd, even, or neither	Section 1.3 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Transformations of Functions	- Identify and graph cubic, square root, reciprocal, step and other piecewise-defined functions - Use reflections, vertical and horizontal shifts to sketch graphs of functions	Section 1.4 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts Geogebra for animations
Combinations of Functions	- Perform computational operations with functions - Find compositions of functions	Section 1.5 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets

		• Kuta (free) for additional problems/handouts
Inverse Functions	• Find inverse functions • Determine if a function is one-to-one	Section 1.6 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets
*Rational Expressions	• Simplify rational expressions (addition/subtraction, multiplication/division) and complex fractions	Supplemental Topic - helpful to include this topic to assist with upcoming rational expression operations in Trigonometry (Proving & Verifying Identities), as well as Limits at the end of the year
Assessments (Benchmark, Formative, Summative, Alternative): Formative: Worksheets/Handouts Formative/Summative: Group work Formative: Ungraded & graded homework assignments Formative: Responses to discussion questions Formative/Summative: Do-Nows/Exit Tickets Summative: Written tests and quizzes Alternative: Projects		
Interdisciplinary Connections: Selected problems throughout the text reflect real-life situations in a variety of careers Overlapping of the use of mathematical concepts in science, engineering, data analysis (STEM applications) Problem Based Learning: Section 1.7 offers extension of content to data and regression analysis (used throughout the unit to supplement)		
Career Readiness, Life Literacies, and Key Skills Standards (21st Century Skills): 9.4.8.CT.2: Develop multiple solutions to a problem and evaluate short- and long-term effects to determine the most plausible option		

Computer Science & Design Thinking Standards (Technology Integration):

8.1.8.DA.1: Organize and transform data collected using computational tools to make it usable for a specific purpose

Climate Change Standards:

8.1.12.DA.1: Create interactive data visualizations using software tools to help others better understand real world phenomena, including climate change

Other Content Standards:**SEL Competencies:**

Responsible Decision-Making: Develop, implement, and model effective problem-solving and critical thinking skills

Self-Management: Identify and apply ways to persevere or overcome barriers through alternative methods to achieve one's goals

UNIT OVERVIEW	
Unit # and Title: II. Functions: Polynomial, Rational, Exponential and Logarithmic	Suggested Time Frame / Pacing: 24 - 30 days
Essential Questions: How do you sketch the graphs of polynomial functions? How do you divide a polynomial by another polynomial to find the zeros of a function? How do you find the domain and asymptotes of a rational function to help sketch its graph? How do you write and graph exponential and logarithmic functions? How do you use properties of exponents and logarithms to solve exponential and logarithmic equations?	
NJSLS Content Standards: MA.A-APR.B.2 - Know and apply the Remainder Theorem: For a polynomial $p(x)$ and a number a , the remainder on division by $x - a$ is $p(a)$, so $p(a) = 0$ if and only if $x - a$ is a factor of $p(x)$. MA.A-APR.D.6 - Rewrite simple rational expressions in different forms; write $\frac{a(x)}{b(x)}$ in the form $q(x) + \frac{r(x)}{b(x)}$, where $a(x)$, $b(x)$, $q(x)$, and $r(x)$ are polynomials with the degree of $r(x)$ less than the degree of $b(x)$ using inspection, long division, or, for the more complicated examples, a computer algebra system. MA.A-APR.D.7 - Understand that rational expressions form a system analogous to the rational numbers, closed under addition, subtraction, multiplication, and division by a nonzero rational expression; add, subtract, multiply, and divide rational expressions. MA.N-CN.A.2 - Use the relation $i^2 = -1$ and the commutative, associative, and distributive properties to add, subtract, and multiply complex numbers. MA.N-CN.A.3 - Find the conjugate of a complex number; use conjugates to find moduli and quotients of complex numbers. MA.N-CN.B.5 - Represent addition, subtraction, multiplication, and conjugation of complex numbers geometrically on the complex plane; use properties of this representation for computation. MA.A-REI.B.4b - Solve quadratic equations by inspection (e.g., for $x^2 = 49$), taking square roots, completing the square, the quadratic formula and factoring, as appropriate to the initial form of the equation. Recognize when the quadratic formula gives complex solutions and write them as $a \pm bi$ for real numbers a and b .	

MA.F-IF.B.5 - Relate the domain of a function to its graph and, where applicable, to the quantitative relationship it describes. For example, if the function gives the number of person-hours it takes to assemble n engines in a factory, then the positive integers would be an appropriate domain for the function.

MA.F-IF.C.7c - Graph polynomial functions, identifying zeros when suitable factorizations are available, and showing end behavior.

MA.F-IF.C.7d - Graph rational functions, identifying zeros and asymptotes when suitable factorizations are available, and showing end behavior.

MA.F-IF.C.7e - Graph exponential and logarithmic functions, showing intercepts and end behavior.

MA.F-IF.C.8b - Use the properties of exponents to interpret expressions for exponential functions.

MA.F-LE.A.2 - Construct linear and exponential functions, including arithmetic and geometric sequences, given a graph, a description of a relationship, or two input-output pairs (include reading these from a table).

MA.F-LE.A.4 - Understand the inverse relationship between exponents and logarithms. For exponential models, express as a logarithm the solution to $ab^{ct} = d$ where a , c , and d are numbers and the base b is 2, 10, or e ; evaluate the logarithm using technology.

MA.F-BF.B.5 - Use the inverse relationship between exponents and logarithms to solve problems involving logarithms and exponents.

Unit Content Vocabulary:

Polynomial function	Quadratic Function	Asymptotes
Minimum and Maximum Values	Power functions	Leading Coefficient Test
Zeros of Polynomial Functions	Multiplicity of Zeros	End-Behavior
Intermediate Value Theorem	The Division Algorithm	The Remainder Theorem
Synthetic division	Complex number	Imaginary number
Pure imaginary number	Standard form	Complex conjugates
Fundamental Theorem of Algebra	Linear Factorization Theorem	Rational Zero Test
Rational function	Domain of a rational function	Nonlinear inequalities
Horizontal and Vertical Asymptotes	Exponential function	Compound interest
Half-life	Logarithmic function	Natural logarithm

Properties of Logarithms e (Euler's number) Exponential and Logarithmic equations		
General Resources: Textbook, Graphing Calculators, Desmos, DeltaMath, Khan Academy, Youtube videos, Kuta Software, Math Medic, Quizizz, Kahoot!, Citizen Math, Which One Doesn't Belong? (wodb), Open Middle, GeoGebra		
Topic and Related Content: <i>What information do students need to know, by topic?</i>	Objectives: <i>Students will be able to:</i>	Suggested Resources, Core and Supplemental Materials and Activities, Technology resources <i>(Include Varied Levels of Text)</i>
Quadratic Functions	- Analyze graphs of quadratic functions - Write quadratic functions in standard form and use the results to sketch graphs of functions - Find maximum and minimum values of quadratic functions in real-life applications	Section 2.1 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Polynomial Functions	- Use transformations to sketch graphs of polynomial functions - Use the Leading Coefficient Test to determine the end behavior of graphs of polynomial functions - Use long or synthetic division to divide polynomials by other polynomials - Use the Remainder Theorem and the Factor Theorem	Section 2.2 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts

Real Zeros of Polynomial Functions	• Find and use zeros of polynomial functions as sketching aids • Use the multiplicity of a zero to aid in sketching the graph • Use the Intermediate Value Theorem to help locate zeros of polynomial functions	Section 2.3 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Complex Numbers	• Use the imaginary unit i to write complex numbers • Add, subtract, multiply complex numbers • Use complex conjugates to write the quotient of two complex numbers in standard form • Find complex solutions of quadratic equations	Section 2.4 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
The Fundamental Theorem of Algebra	• Use the Fundamental Theorem of Algebra to determine the number of zeros of polynomial functions • Find rational zeros of polynomial functions • Find conjugate pairs of complex zeros	Section 2.5 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets

	Find zeros of polynomials by factoring	Kuta (free) for additional problems/handouts
Rational Functions & Asymptotes	- Find the domains of rational functions - Find the vertical and horizontal asymptotes of graphs of rational functions - Locate “holes” in a rational function	Section 2.6 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Graphs of Rational Functions	- Analyze and sketch the graphs of rational functions - Use rational functions to model and solve real-life problems	Section 2.7 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Exponential Functions	- Recognize, evaluate and graph exponential functions with base a - Recognize, evaluate and graph exponential functions with base e	Section 3.1 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization

	• Determine domain & range for exponential functions • Use exponential functions to model and solve real life problems	• Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Logarithmic Functions	• Recognize, evaluate and graph logarithmic functions with base a • Recognize, evaluate and graph logarithmic functions with base e • Determine domain & range for logarithmic functions • Use logarithmic functions to model and solve real life problems	Section 3.2 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Properties of Logarithms	• Use the change-of-base formula to rewrite and evaluate logarithmic expressions • Use properties of logarithms to evaluate, rewrite, expand or condense logarithmic expressions	Section 3.3 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts

Solving Exponential and Logarithmic Equations	Solve simple and complicated exponential and logarithmic equations	Section 3.4 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
*Exponential & Logarithmic Models	Use exponential and logarithmic equations to model and solve real-life problems	Section 3.5 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition STEM project: Loan Amortization (Car Payments, College Loans, Mortgage Payments) for Budgets Application Problems relating to rising temperature (8.1.12.DA.1)
Assessments (Benchmark, Formative, Summative, Alternative): Formative: Worksheets/Handouts Formative/Summative: Group work Formative: Ungraded & graded homework assignments Formative: Responses to discussion questions Formative/Summative: Do-Nows/Exit Tickets Summative: Written tests and quizzes Alternative: Projects		

Interdisciplinary Connections:

Selected problems throughout the text reflect real-life situations in a variety of careers
Overlapping of the use of mathematical concepts in science, engineering, data analysis (STEM applications)
Problem Based Learning: Population Growth, Compound Interest, Richter Scale
STEM project: Loan Amortization (Car Payments, College Loans, Mortgage Payments) for Budgets

Career Readiness, Life Literacies, and Key Skills Standards (21st Century Skills):

9.4.8.CT.2: Develop multiple solutions to a problem and evaluate short- and long-term effects to determine the most plausible option

Computer Science & Design Thinking Standards (Technology Integration):

8.1.8.DA.1: Organize and transform data collected using computational tools to make it usable for a specific purpose

Climate Change Standards:

8.1.12.DA.1: Create interactive data visualizations using software tools to help others better understand real world phenomena, including climate change

Other Content Standards:**SEL Competencies:**

Responsible Decision-Making: Develop, implement, and model effective problem-solving and critical thinking skills

Self-Management: Identify and apply ways to persevere or overcome barriers through alternative methods to achieve one's goals

UNIT OVERVIEW	
Unit # and Title:	Suggested Time Frame / Pacing:
III. Trigonometry and Related Topics	65 - 75 days
Essential Questions: How do you describe angles and angular movement? How do you evaluate trigonometric functions by using the unit circle? How do you use trigonometry to find unknown side lengths and angle measures in triangles? How do you sketch the graphs of the six trigonometric functions? How do you rewrite trigonometric expressions in order to simplify and evaluate trigonometric expressions, and solve trigonometric equations? How do you represent and perform operations on complex numbers by using trigonometry?	
NJSLS Content Standards: MA.F-TF.A.1 - Understand radian measure of an angle as the length of the arc on the unit circle subtended by the angle. MA.F-TF.A.2 - Explain how the unit circle in the coordinate plane enables the extension of trigonometric functions to all real numbers, interpreted as radian measures of angles traversed counterclockwise around the unit circle. MA.F-TF.A.3 - Use special triangles to determine geometrically the values of sine, cosine, tangent for $\frac{\pi}{3}$, $\frac{\pi}{4}$, and $\frac{\pi}{6}$ and use the unit circle to express the values of sine, cosines, and tangent for $\pi - x$, $\pi + x$ and $2\pi - x$ in terms of their values for x , where x is any real number. MA.F-TF.B.5 - Choose trigonometric functions to model periodic phenomena with specified amplitude, frequency, and midline. MA.F-TF.C.8 - Prove the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$ and use it to find $\sin\theta$, $\cos\theta$, or $\tan\theta$ given $\sin\theta$, $\cos\theta$, or $\tan\theta$ and the quadrant of the angle. MA.F-TF.C.9 - Prove the addition and subtraction formulas for sine, cosine, and tangent and use them to solve problems. MA.G-SRT.C.6 - Understand that by similarity, side ratios in right triangles are properties of the angles in the triangle, leading to definitions of trigonometric ratios for acute angles.	

MA.G-SRT.C.8 - Use trigonometric ratios and the Pythagorean Theorem to solve right triangles in applied problems.

MA.G-SRT.D.11 - Understand and apply the Law of Sines and the Law of Cosines to find unknown measurements in right and non-right triangles (e.g., surveying problems, resultant forces).

MA.N-CN.B.4 - Represent complex numbers on the complex plane in rectangular and polar form (including real and imaginary numbers), and explain why the rectangular and polar forms of a given complex number represent the same number.

MA.N-CN.B.5 - Represent addition, subtraction, multiplication, and conjugation of complex numbers geometrically on the complex plane; use properties of this representation for computation.

MA.N-CN.B.6 - Calculate the distance between numbers in the complex plane as the modulus of the difference, and the midpoint of a segment as the average of the numbers at its endpoints.

Unit Content Vocabulary:

Trigonometry	angle	initial side	vertex
terminal side	standard position	positive angles	negative angles
co-terminal	central angle	radian	complementary angles
supplementary angles	linear speed	angular speed	sector
unit circle	sine	cosine	tangent
secant	cosecant	cotangent	period of f
co-functions	trigonometric identity	amplitude	vertical shrinking
vertical stretch	asymptote	damping factor	inverse trig functions
harmonic motion	Pythagorean identity	multiple angle formulas	half-angle formula
sum and difference formulas	Law of Sines	Law of Cosines	SAS Area formula
SSS Area (Heron's) Formula	trig form of complex #	DeMoivre's Theorem	absolute value of complex #

General Resources: Textbook, Graphing Calculators, Desmos, DeltaMath, Khan Academy, Youtube videos, Kuta Software, Math Medic, Quizizz, Kahoot!, Citizen Math, Which One Doesn't Belong? (wodb), Open Middle, GeoGebra

Topic and Related Content:

What information do students need to know, by topic?

Objectives:

Students will be able to:

Suggested Resources, Core and Supplemental Materials and Activities, Technology resources
(Include Varied Levels of Text)

Radian & Degree Measure

- Understand radian measure for measuring angles

Section 4.1
of Precalculus with Limits: A Graphing Approach

	• Convert radians to degrees and vice-versa • Sketch an angle in standard position • Find coterminal angles for a given angle	by Larson & Battaglia, 8th edition • Textbook Practice • Geogebra for animations • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Trigonometric Functions: The Unit Circle	• Understand the value of the unit circle in evaluating trigonometric functions of any angle • Understand how trigonometric ratios are developed from right angles	Section 4.2 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • Geogebra for animations • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Right Triangle Trigonometry	• Understand how trigonometric ratios are developed from right angles	Section 4.3 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Trigonometric Functions of Any Angle	• Understand how to evaluate trigonometric functions of any angle • Understand how trigonometric ratios are developed from right angles	Section 4.4 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • Geogebra for animations • DeltaMath Practice

		• Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Graphs of Sine & Cosine Functions	• Understand the periodicity, amplitude, and symmetry in graphing trigonometric functions • Understand how to graph the sine & cosine functions	Section 4.5 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • Geogebra for animations • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts STEM project: Sinusoidal Applications (Tides, Hours of Daylight, Ferris Wheels, Orbits/Arc Length)
Graphs of Other Trigonometric Functions	• Understand how to graph the six trigonometric functions	Section 4.6 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Inverse Trigonometric Functions	• Understand how to evaluate inverse trigonometric functions	Section 4.7 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice

		• Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Using Fundamental Identities	• Understand algebraic manipulation of trigonometric identities	Section 5.1 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Verifying Trigonometric Identities	• Prove & verify trigonometric identities, through algebraic manipulation of trigonometric expressions	Section 5.2 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Solving Trigonometric Equations	• Use identities and algebraic manipulation to solve a variety of trigonometric equations	Section 5.3 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets

		• Kuta (free) for additional problems/handouts
Sum and Difference Formulas	• Understand how to apply the sum and difference formulas	Section 5.4 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Multiple Angle and Product-to-Sum Formulas	• Understand how to apply the double angle and half-angle formulas	Section 5.5 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Law of Sines	• Use the Law of Sines to solve an oblique triangle • Calculate the area of a triangle with the SAS Area formula	Section 6.1 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • Geogebra for animations (ambiguous case) • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Law of Cosines	• Use the Law of Cosines to solve an oblique triangle	Section 6.2 <i>of Precalculus with Limits: A Graphing Approach</i>

	Calculate the area of a triangle with the SSS Area formula (Heron's Formula)	by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
*The Complex Plane	- Convert a complex number from standard form to trigonometric (or polar) form, and vice-versa - Plot a complex number on the complex plane	Section 6.5 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
*Trigonometric Form of a Complex Number	- Multiply & divide complex numbers in trigonometric form - Use DeMoivre's Theorem	Section 6.6 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Assessments (Benchmark, Formative, Summative, Alternative): Formative: Worksheets/Handouts Formative/Summative: Group work Formative: Ungraded & graded homework assignments Formative: Responses to discussion questions Formative/Summative: Do-Nows/Exit Tickets Summative: Written tests and quizzes Alternative: Projects		
Interdisciplinary Connections: Selected problems throughout the text reflect real-life situations in a variety of careers Overlapping of the use of mathematical concepts in science, engineering, data analysis (STEM applications)		

Problem Based Learning: Periodic Phenomena in Nature (low/high tide, hours of daylight), Right Triangle Applications (angle of elevation/depression), Circular Motion (arc length, ferris wheel movement, latitude/longitude, linear/angular speed), Navigation & bearing problems, Distances between objects using triangulation, harmonic motion

STEM project: Sinusoidal Applications (Tides, Hours of Daylight, Ferris Wheels, Orbits/Arc Length)

Career Readiness, Life Literacies, and Key Skills Standards (21st Century Skills):

9.4.8.CT.2: Develop multiple solutions to a problem and evaluate short- and long-term effects to determine the most plausible option

Computer Science & Design Thinking Standards (Technology Integration):

8.1.8.DA.1: Organize and transform data collected using computational tools to make it usable for a specific purpose

8.2.2.A.1: Define products produced as a result of technology or of nature

Climate Change Standards:

8.1.12.DA.1: Create interactive data visualizations using software tools to help others better understand real world phenomena, including climate change

Other Content Standards:

NJSEL Competencies:

Responsible Decision-Making: Develop, implement, and model effective problem-solving and critical thinking skills

Self-Management: Identify and apply ways to persevere or overcome barriers through alternative methods to achieve one's goals

UNIT OVERVIEW

Unit # and Title:	Suggested Time Frame / Pacing:
IV. Sequences, Series, and Probability	20 - 30 days
Essential Questions: How do you represent a sequence of numbers or the sum of a sequence? How do you find the n th term or partial sum of an arithmetic or geometric sequence? How do you find the expansion of a binomial $(x + y)^n$? How do you count the number of ways in which an event can occur? How do you find the probability that a series of events will occur?	

NJSLS Content Standards:

MA.F-IF.A.3 - Recognize that sequences are functions, sometimes defined recursively, whose domain is a subset of the integers.

MA.F-BF.A.2 - Write arithmetic and geometric sequences both recursively and with an explicit formula, use them to model situations, and translate between the two forms.

MA.A-APR.C.5 - Know and apply the Binomial Theorem for the expansion of $(x + y)^n$ in powers of x and y for a positive integer n , where x and y are any numbers, with coefficients determined for example by Pascal's Triangle.

MA.S-CP.B.9 - Use permutations and combinations to compute probabilities of compound events and solve problems.

Unit Content Vocabulary:

sequence	finite sequence	infinite sequence
terms of the sequence	explicit definition	recursive definition
factorial	Fibonacci Sequence	series
finite series	infinite series	arithmetic sequence
geometric sequence	common difference	Binomial Theorem
Pascal's Triangle	convergent series	divergent series
Fundamental Counting Principle	Permutation	Combination
probability of an event	independent events	mutually exclusive

General Resources: Textbook, Graphing Calculators, Desmos, DeltaMath, Khan Academy, Youtube videos, Kuta Software, Math Medic, Quizizz, Kahoot!, Citizen Math, Which One Doesn't Belong? (wodb), Open Middle, GeoGebra

Topic and Related Content:

What information do students need to know, by topic?

Objectives:

Students will be able to:

Suggested Resources, Core and Supplemental Materials and Activities, Technology resources
(Include Varied Levels of Text)

Sequences & Series

- Use sequence notation to write the terms of sequences
- Use factorial notation
- Use summation notation to write sums

Section 8.1

of *Precalculus with Limits: A Graphing Approach* by Larson & Battaglia, 8th edition

- Textbook Practice
- [DeltaMath](#) Practice

	Find the sums of series	Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Arithmetic Sequences and Partial Sums	- Recognize, write and find the nth terms of arithmetic and geometric sequences - Find nth partial sums of arithmetic sequences - Use arithmetic sequences to model and solve real life problems	Section 8.2 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Geometric Sequences and Series	- Use geometric sequences to model and solve real life problems - Find the sum of a finite geometric sequence - Find the sum of an infinite geometric series	Section 8.3 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
The Binomial Theorem	- Use the Binomial Theorem to calculate binomial coefficients - Use Pascal's Triangle to expand binomials	Section 8.4 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice

		• Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Counting Principles	• Solve simple counting problems • Use the Fundamental Counting Principle to solve counting problems • Use permutations to solve counting problems • Use combinations to solve counting problems.	Section 8.5 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Probability	• Find the probabilities of events • Find the probabilities of mutually exclusive events • Find the probabilities of independent events • Find the probability of the complement on an event	Section 8.6 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Assessments (Benchmark, Formative, Summative, Alternative): Formative: Worksheets/Handouts Formative/Summative: Group work Formative: Ungraded & graded homework assignments Formative: Responses to discussion questions Formative/Summative: Do-Nows/Exit Tickets Summative: Written tests and quizzes		

Alternative: Projects

Interdisciplinary Connections:

Selected problems throughout the text reflect real-life situations in a variety of careers

Overlapping of the use of mathematical concepts in science, engineering, data analysis (STEM applications)

Problem Based Learning: actuarial sciences, statisticians, sports probability, odds, lotteries, forensics

Career Readiness, Life Literacies, and Key Skills Standards (21st Century Skills):

9.4.8.CT.2: Develop multiple solutions to a problem and evaluate short- and long-term effects to determine the most plausible option

Computer Science & Design Thinking Standards (Technology Integration):

8.1.8.DA.1: Organize and transform data collected using computational tools to make it usable for a specific purpose

8.2.2.A.1: Define products produced as a result of technology or of nature

Climate Change Standards:

N/A

Other Content Standards:

SEL Competencies:

Responsible Decision-Making: Develop, implement, and model effective problem-solving and critical thinking skills

Self-Management: Identify and apply ways to persevere or overcome barriers through alternative methods to achieve one's goals

UNIT OVERVIEW																				
Unit # and Title: V. Conics, Parametrics, and Polar Equations* (HONORS only)	Suggested Time Frame / Pacing: 10 - 14 days																			
Essential Questions: How do you recognize each conic section and apply its properties to solve problems? How do you write (parametric) equations to describe the motion of a point in a plane? How do you describe the position of a point in a plane using distance and angle rather than x - and y - coordinates? How do you sketch graphs of polar equations?																				
NJSLS Content Standards: MA.G-GPE.A.1 - Derive the equation of a circle of given center and radius using the Pythagorean Theorem; complete the square to find the center and radius of a circle given by an equation. MA.G-GPE.A.2 - Derive the equation of a parabola given a focus and directrix. MA.G-GPE.A.3 - Derive the equations of ellipses and hyperbolas given the foci, using the fact that the sum or difference of distances from the foci is constant. MA.F-IF.B.4 - For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.																				
Unit Content Vocabulary: <table><tr><td>conic sections</td><td>circle</td><td>parabola</td></tr><tr><td>ellipse</td><td>hyperbola</td><td>parametric equations</td></tr><tr><td>parameter</td><td>rectangular form</td><td>rectangular coordinates</td></tr><tr><td>polar coordinates</td><td>polar coordinate system</td><td>pole</td></tr><tr><td>polar graphs</td><td>limaçon</td><td>cardioid</td></tr><tr><td>inner loop limaçon</td><td>dimpled limaçon</td><td>rose curve</td></tr></table>			conic sections	circle	parabola	ellipse	hyperbola	parametric equations	parameter	rectangular form	rectangular coordinates	polar coordinates	polar coordinate system	pole	polar graphs	limaçon	cardioid	inner loop limaçon	dimpled limaçon	rose curve
conic sections	circle	parabola																		
ellipse	hyperbola	parametric equations																		
parameter	rectangular form	rectangular coordinates																		
polar coordinates	polar coordinate system	pole																		
polar graphs	limaçon	cardioid																		
inner loop limaçon	dimpled limaçon	rose curve																		

General Resources: Textbook, Graphing Calculators, Desmos, DeltaMath, Khan Academy, Youtube videos, Kuta Software, Math Medic, Quizizz, Kahoot!, Citizen Math, Which One Doesn't Belong? (wodb), Open Middle, GeoGebra

Topic and Related Content: <i>What information do students need to know, by topic?</i>	Objectives: <i>Students will be able to:</i>	Suggested Resources, Core and Supplemental Materials and Activities, Technology resources <i>(Include Varied Levels of Text)</i>
*Circles & Parabolas	• Write the equation of a parabola in standard form. • Write the equation of a circle in standard form.	Section 9.1 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Ellipses	• Write the equation of an ellipse in standard form.	Section 9.2 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Hyperbolas	• Write the equation of a hyperbola in standard form.	Section 9.3 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization

		• Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Parametrics	• Evaluate and graph sets of parametric equations for given values of the parameter. • Rewrite sets of parametric equations as single rectangular equations by eliminating the parameter.	Section 9.4 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Polar Coordinates	• Plot points on the polar coordinate system. • Convert points from rectangular to polar form and vice versa.	Section 9.5 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Polar Graphs	• Graph polar equations by point plotting.	Section 9.6 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition • Textbook Practice • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Assessments (Benchmark, Formative, Summative, Alternative):		

Formative: Worksheets/Handouts
 Formative/Summative: Group work
 Formative: Ungraded & graded homework assignments
 Formative: Responses to discussion questions
 Formative/Summative: Do-Nows/Exit Tickets
 Summative: Written tests and quizzes
 Alternative: Projects

Interdisciplinary Connections:

Selected problems throughout the text reflect real-life situations in a variety of careers
 Overlapping of the use of mathematical concepts in science, engineering, data analysis (STEM applications)
 Problem Based Learning: physics/motion problems, satellite dishes, planetary orbits, mirrors, polar microphones/sound engineering

Career Readiness, Life Literacies, and Key Skills Standards (21st Century Skills):

9.4.8.CT.2: Develop multiple solutions to a problem and evaluate short- and long-term effects to determine the most plausible option

Computer Science & Design Thinking Standards (Technology Integration):

8.1.8.DA.1: Organize and transform data collected using computational tools to make it usable for a specific purpose

Climate Change Standards:

N/A

Other Content Standards:

NJSEL Competencies:

Responsible Decision-Making: Develop, implement, and model effective problem-solving and critical thinking skills

Self-Management: Identify and apply ways to persevere or overcome barriers through alternative methods to achieve one's goals

UNIT OVERVIEW	
Unit # and Title:	Suggested Time Frame / Pacing:
VI. Introduction to Calculus: Limits & Derivatives	20 - 35 days
Essential Questions: How do you find and interpret the limit of a function for a certain value of x ? How do you evaluate limits that cannot be solved through use of direct substitution? How do you find the slope of a graph at any single point? How do you find the limits of functions at infinity? What is the derivative of a function? How do you use the Power, Product, and Quotient rules to find derivatives? How do you find the derivatives of sine and cosine functions?	
NJSLS Content Standards: MA.F-IF.B.4 - For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity. MA.F-IF.B.6 - Calculate and interpret the average rate of change of a function (presented symbolically or as a table) over a specified interval. Estimate the rate of change from a graph. MA.F-BF.B.3 - Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$ and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them. MA.F-IF.C.7c - Graph polynomial functions, identifying zeros when suitable factorizations are available, and showing end behavior. MA.F-IF.C.7d - Graph rational functions, identifying zeros and asymptotes when suitable factorizations are available, and showing end behavior. MA.F-IF.C.7e - Graph exponential and logarithmic functions, showing intercepts and end behavior.	

Unit Content Vocabulary: limit end-behavior slope average rate of change Power Rule Higher-Order derivatives indeterminate form infinity tangent line instantaneous rate of change Product Rule increasing/decreasing asymptote unbounded secant line derivative Quotient Rule		
General Resources: Textbook, Graphing Calculators, Desmos, DeltaMath, Khan Academy, Youtube videos, Kuta Software, Math Medic, Quizizz, Kahoot!, Citizen Math, Which One Doesn't Belong? (wodb), Open Middle, GeoGebra		
Topic and Related Content: <i>What information do students need to know, by topic?</i>	Objectives: <i>Students will be able to:</i>	Suggested Resources, Core and Supplemental Materials and Activities, Technology resources <i>(Include Varied Levels of Text)</i>
Introduction to Limits	Understand what a limit is and be able to calculate a limit from a graph.	Section 11.1 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
Techniques for Evaluating Limits	- Evaluate a limit algebraically, using a variety of techniques. - Evaluate a limit of the difference quotient. - Sketch the graph of a piecewise function in order to evaluate its limit.	Section 11.2 <i>of Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets

	Evaluate a one-sided limit.	Kuta (free) for additional problems/handouts
*The Tangent Line Problem	- Use limits to find the slope of a graph at a given point. - Use limits to find the derivative of a function. - Use the derivative to find the slope of a line tangent to a function at a given point.	Section 11.3 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice Geogebra for animations (secant lines approaching tangent line) DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
*Limits at Infinity	Understand what a limit at infinity is and be able to calculate it from an equation.	Section 11.4 of <i>Precalculus with Limits: A Graphing Approach</i> by Larson & Battaglia, 8th edition - Textbook Practice DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts
*Derivatives: Power Rule	- Find the derivative of a function using the Power Rule. - Write the equation of a line tangent to a function at a given x-value. - Find where the graph of a function has horizontal tangent lines.	From a Calculus Textbook - Textbook Practice (photocopy problem set) DeltaMath Practice Desmos graphing visualization Quizizz & Kahoot! for Do Nows/Exit Tickets Kuta (free) for additional problems/handouts

	• Determine the average rate of change for a function over a closed interval. • Determine the instantaneous rate of change for a function at a point.	
*Derivatives: Sine & Cosine	• Find the derivative of sine and cosine functions.	From a Calculus Textbook • Textbook Practice (photocopy problem set) • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Derivatives: Product & Quotient Rules	• Use the Product and Quotient Rules to find the derivative of a function.	From a Calculus Textbook • Textbook Practice (photocopy problem set) • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
*Derivatives: Higher Order Derivatives	• Find higher order derivatives. • Determine concavity of a graph using 2nd derivatives.	From a Calculus Textbook • Textbook Practice (photocopy problem set) • DeltaMath Practice • Desmos graphing visualization • Quizizz & Kahoot! for Do Nows/Exit Tickets • Kuta (free) for additional problems/handouts
Assessments (Benchmark, Formative, Summative, Alternative):		

Formative: Worksheets/Handouts
 Formative/Summative: Group work
 Formative: Ungraded & graded homework assignments
 Formative: Responses to discussion questions
 Formative/Summative: Do-Nows/Exit Tickets
 Summative: Written tests and quizzes
 Alternative: Projects

Interdisciplinary Connections:

Selected problems throughout the text reflect real-life situations in a variety of careers
 Overlapping of the use of mathematical concepts in science, engineering, data analysis (STEM applications)
 Problem Based Learning: Engineering, Physics free-fall problems (position/velocity/acceleration), EZ-Pass analysis, Pneumatic Road Tubes, graphing calculator calculus features

Career Readiness, Life Literacies, and Key Skills Standards (21st Century Skills):

9.4.8.CT.2: Develop multiple solutions to a problem and evaluate short- and long-term effects to determine the most plausible option

Computer Science & Design Thinking Standards (Technology Integration):

8.1.8.DA.1: Organize and transform data collected using computational tools to make it usable for a specific purpose

Climate Change Standards:

N/A

Other Content Standards:

SEL Competencies:

Responsible Decision-Making: Develop, implement, and model effective problem-solving and critical thinking skills

Self-Management: Identify and apply ways to persevere or overcome barriers through alternative methods to achieve one's goals

Career Education Integration: *(Provides opportunity to acquire information about career interests, aligned to 9.2 Career Awareness and Planning)*

9.2.12.CAP.2: Develop college and career readiness skills by participating in opportunities such as structured learning experiences, apprenticeships, and dual enrollment programs.

(*Honors Precalculus is offered as a dual-credit option with Middlesex College.)

Career connections: Various fields of engineering (civil, electrical, mechanical).

Modifications: See district guide on accommodations and modifications for select groups

Special Education / 504	Multilingual Learners	At-Risk Of Failure	Gifted And Talented
Modifications and accommodations for students with IEPs or 504's are designed and documented in daily lesson plans	Modifications and accommodations for ML students are designed and documented in daily lesson plans	Interventions and strategies to support at-risk students are designed and documented in daily lesson plans	Enrichment and acceleration opportunities for G&T students are designed and documented in daily lesson plans

Additional Curricular Mandates: *Note how the below requirements are addressed in the grade level curriculum.*

Diversity & Inclusion:	N/A
Holocaust and genocides:	N/A
Amistad Law- History and contributions of African-Americans	N/A
Asian American, Pacific Islander history and contributions: (Social Studies)	N/A

History of Disabled and LGBTQ+ Persons (Middle and High School):	N/A
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